Case 45: WU-E heat-to-urban rate-of-spread response

1 Verification purpose

This manufactured coupling test verifies the Wildland–Urban ELMFIRE (WU–E) model's conversion of heat received by an urban cell into its local level-set rate of spread (ROS). It exercises the heat-driven building-spread formulation and subsequent construction of the directional spread velocity. The intended reference calculation uses a fixed critical material fire-thermal property (FTP); it is sensitive to energy or unit normalization, ellipse partitioning, normal-vector orientation, an artificial 30,000-kJ or 35-mph discontinuity, and failure to honor the material table. The source revision reviewed for the verification specification is `a2dfbcdf72209733c000e5d3431e44723e15efea`.

The five reviewed model components were identical to that source revision. The simulation, output, and initialization procedures contained other changes outside the reviewed components. Consequently, the design review establishes the provenance of the reference formulation, not the identity of every component in a subsequently supplied ELMFIRE version.

One burning urban source and one adjacent, initially positive urban receiver form the measured pair in a nonburnable domain. Four disconnected urban guard cells keep the tagged-list termination guard inactive but remain far from the front and are not observables. Per-step energy, wind, receiver normal, design-fire amplitude, and table critical material fire-thermal property are varied; spotting, crown fire, terrain, moisture gradients, and stochastic forcing are disabled. This is implementation verification, not evidence that the empirical response is physically valid. The verification question is: for all 12 simulation conditions, does the executable's receiver-cell spread-rate raster agree within 0.005 relative error with the independently recomputed mapping driven by its own recorded direct flame contact (DFC)-plus-radiation stimulus, while also satisfying the continuity, directional, amplitude, and FTP-response comparisons?

Current result: PASS. 12 of 12 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

At one receiver the executable records transient heat fluxes q''_{DFC} and q''_{rad} in kW m^{-2}. For timestep Δt and square cell width Δx, the downstream routine forms

$$E = (q''_{\mathrm{DFC}} + q''_{\mathrm{rad}})\Delta t \Delta x^2, \qquad q_A = E/\Delta x^2.$$

Because kilowatts are kilojoules per second, E is in kilojoules. The intended downscaled model uses the material value $F_{TP,crit}$ without an energy-dependent replacement or wind-speed divisor. The absolute velocity in feet per minute is

$$U = \min\left(10^5, \frac{60E}{0.3048\,\Delta t\,\Delta x\,F_{TP,crit}}\right) = \min\left(10^5, \frac{196.85\,q''\Delta x}{F_{TP,crit}}\right).$$

The nominal default is $F_{TP,crit} = 5400$ kJ m^{-2}; the case also uses 3000 and 6000 to verify the inverse response. The peer-reviewed papers describe the accumulated-heat/FTP ignition and heat-derived urban spread concept [1, 2]. The pinned source instead replaces FTP below/above 30,000 kJ and applies a 1.8 divisor at and below 35 mph. Those operations are documented deviations, not acceptance equations.

The UCB ellipse supplies major radius a, minor radius b, and focus offset e. The three response weights are

$$f_h = a + e, \qquad f_s = 2b, \qquad f_b = a - e,$$

so the nominal head, side, and back velocities are $V_i = U f_i/(f_h + f_s + f_b)$. The case-specific evaluator then applies the same closed ellipse-normal construction, rather than treating these three labels as three scalar shortcuts. In full, for front normal $\boldsymbol{n} = (n_x, n_y)$ and direction-of-maximum-spread unit vector $\boldsymbol{s} = (s_x, s_y)$, the implementation evaluates

$$c = n_x s_x + n_y s_y, \qquad s = n_x s_y - n_y s_x,$$

$$L = \begin{cases} \min((V_h + V_b)/(2V_s), 10)\,, & V_s > 10^{-4}, \\ 1, & V_s \le 10^{-4}, \end{cases}$$

$$A = \max\left(\frac{V_h + V_b}{2}, 10^{-10}\right), \qquad B = \frac{1}{2}\max\left(\frac{V_h + V_b}{L}, 10^{-10}\right),$$

$$D = \max\left(\sqrt{A^2c^2 + B^2s^2}, 10^{-10}\right), \quad v_{\parallel} = \frac{A^2 c}{D} + \frac{V_h - V_b}{2}, \quad v_{\perp} = \frac{B^2 s}{D},$$

$$v_x = v_{\parallel} s_x + v_{\perp} s_y, \qquad v_y = v_{\parallel} s_y - v_{\perp} s_x, \qquad V = \sqrt{v_x^2 + v_y^2}.$$

The flat grid makes the source projection factors unity. Wind is from 270°, hence $\boldsymbol{s} = (1, 0)$. The manufactured target normals are east (head), west (back), and north (side); raster rows run north to south while the stored mathematical normal is expressed in east/north coordinates.

The upstream heat values are recorded stimuli for this case; CASE43–44 verify their governing equations. Here characteristic building dimension=20 denotes the source's linear characteristic building dimension in metres, not an area in square metres. The input generator chooses design-fire amplitudes deterministically, but the downstream reference calculation always uses the heat actually written by ELMFIRE. For traceability, the pinned source currently uses the following non-normative normalization while constructing that stimulus:

$$C_H = \frac{\Delta x^2}{\pi (ca)(cb)}, \qquad c = 0.5.$$

For target distance R and directional ellipse distance R_e, the input generator transcribes the overlap and radiation weights

$$C_D = \mathrm{clip}_{[0,1]}\left(\frac{R_e + \Delta x/2 - R}{\Delta x}\right), \qquad C_R = (1 - C_D)\,\mathrm{clip}_{[0,1]}\left(\frac{R_e + 100 + \Delta x/2 - R}{\Delta x}\right).$$

With the implementation's piecewise effective radiation distance R_{eff}, emissivity $\varepsilon = 0.89$, and radiative fraction 0.3, the planned plateau heat is

$$q'' = C_H H_p \left[C_D + \frac{0.3\,\varepsilon C_R \Delta x^2}{4\pi R_{\mathrm{eff}}^2}\right].$$

The generator solves this relation separately for each normal to set the input amplitude. The downstream decision never substitutes planned heat for the recorded heat.

3 Derivation of expected behavior

The energy triplet is constructed at 29900, 30000, and 30100 kJ. The generator uses an explicit IEEE-754 binary32 transcription of the reviewed default-single-precision arithmetic operation sequence to select decimal table amplitudes; the planned realization is exact at all three values, including equality, and is recorded in each simulation specification. Executable heat output remains required to establish the

realized stimulus. With fixed FTP, local speed divided by recorded energy must be continuous across 30,000 kJ. Likewise, the 34.9, 35.0, and 35.1-mph cases include the small physical ellipse change but no artificial divisor jump. A discontinuity at either boundary identifies the known source-only branches.

The terminal spread-rate raster is populated only from the set of cells recorded as burned, so an arbitrary positive receiver would not be observable. The prepared receiver value is instead $\phi_r = 0.001 > 0$. Both source raster readers unconditionally add a perturbation $10^{-3}(0.5 - R)$, with $R \in [0, 1)$; wind-fluctuation settings do not disable it. Thus the receiver remains strictly positive, between 0.0005 and 0.0015, before propagation. The former 2^{-20} value was smaller than this perturbation and could not guarantee initial state. For the side receiver, its five-cell tangential PHI stencil is flat at ϕ_r, removing an artificial tangential local minimum. These four extra stencil cells remain nonburnable and initially positive.

The crossing estimate uses $\boldsymbol{v} \cdot \nabla\phi$, not speed magnitude times a one-dimensional gradient. Its source-facing Superbee gradient is $G = 2(\phi_r - \phi_s)/\Delta x = 0.0451$ per metre. With $\epsilon = 0.0005$, the generator checks the conservative first-stage bound

$$\tfrac{1}{2}\Delta t\,[v_n(G - 4\epsilon/\Delta x) - U(4\epsilon/\Delta x)] > \phi_r + \epsilon,$$

where v_n is normal velocity and U bounds the velocity magnitude, both in metres per second. The bound covers input-noise gradients and discards second-stage inward transport. It is an intended-law setup check; actual arrival remains independently required. All twelve receivers must enter the set of cells recorded as burned at exactly 1 s, with a finite terminal speed. Four positive guard cells keep more than two nodes tagged, and a scheduled ignition at 5 s—strictly after the 4 s stop—keeps the first-step stall predicate false without ever altering the simulated state.

At fixed geometry, doubling versus halving H_p gives a heat-flux ratio of four. The two critical material fire-thermal property simulation conditions use 3000 and 6000 kJ m^{-2} and are otherwise equivalent. Their heat fields must match, while their spread speeds must have the inverse FTP ratio $V_{3000}/V_{6000} = 2$. The equal-energy condition with 35-mph wind, the back-facing receiver, and the side-facing receiver simulation conditions share 35-mph wind and the same designed received energy; they are the matched head, back, and side comparison, leaving only the prescribed receiver normal to select the directional response.

This is a recorded-input reference calculation: the emitted/received heat field at the exact receiver-arrival dump is a measured stimulus, while only the downstream heat-to-ROS calculation is the object of the numerical comparison. It therefore avoids claiming that one executable output independently verifies another upstream heat calculation. Ellipse axes, E as held in working storage, F, absolute U, component velocities, and ellipse length-to-width ratio are internal-only quantities. They are calculated by the evaluator but cannot be asserted as separately observed executable outputs.

4 Verification metrics and acceptance criteria

Before evaluation, the simulation settings, spatial inputs, fuel properties, reference calculations, and analysis method must match those associated with the completed simulation. Their recorded SHA-256 identities are checked, together with the full simulation record and model results. All 12 simulation conditions must have completed using the same identified ELMFIRE program. Any subsequent change to this evidence invalidates the comparison.

With output at every timestep, the exact ledger is dump indices 1–5 at times 0, 1, 2, 3, and 4 s, with only index 5 final. Transient heat fields are identified by output index, whereas terminal arrival-time and spread-rate fields are identified by physical time, 4 s. The final output index 5 is not interpreted as an elapsed time. Evaluation selects DFC plus radiation at the unique dump whose time equals the

receiver's required 1-s time of arrival (TOA), and compares the resulting independent local-speed reference calculation with the terminal receiver speed captured when that cell entered the set of cells recorded as burned. The exact-stop transient heat values must be reset to zero. A missing or stale marker, input identity record or snapshot mismatch, nodata, nonfinite, missing, duplicate, stale, or off-time supporting result makes the case NOT EVALUABLE.

Property	Observable and metric	Selection rationale	Acceptance basis
Heat-to-ROS mapping	maximum relative receiver-speed error	Tests the downstream formula using recorded heat	≤ 0.005, allowing float raster serialization
Receiver observability	TOA and finite terminal speed for every receiver	Proves the compared value was serialized from the set of cells recorded as burned	TOA exactly 1 s within 10^{-6} s; all speeds finite
Fixed-FTP energy continuity	normalized speed for 29900/30000/30100	Detects the non-normative source energy branch	spread ≤ 0.005; energy within 0.2% of design
Wind continuity	errors for 34.9/35.0/35.1 mph	Detects the non-normative 35-mph divisor	all speed errors ≤ 0.005
heat-release rate (HRR) amplitude	recorded double/half heat ratio	Direct metamorphic check with no assumed absolute output	4.00 ± 0.04
Directional mapping	maximum error for the matched 35-mph head/back/side set	Detects orientation and ellipse partition defects	≤ 0.005
critical material fire-thermal property inverse response	speed ratio for 3000/6000	Requires the material table to control spread	2.00 ± 0.01

PASS requires complete evidence and every row to pass. Any calculated breach is FAIL; absent or ambiguous evidence is NOT EVALUABLE and never a pass.

5 Simulation configuration and predicted outcome

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Parameter	Configured value	Units	Isolation role

Grid	25×25, EPSG:32610	–	source well away from boundary
Cell width	20	m	fixed energy normalization and CFL-safe local response
Fuel geometry	source, adjacent receiver, four diagonal guard cells are fire behavior fuel model (FBFM) 91; FBFM 93 elsewhere	category	observable pair plus nonburning termination guards
Initial ϕ	source -0.45, receiver 0.001 (side tangential stencil also 0.001), guards/background (+1)	–	first-heated-step receiver crossing with guards far from the front
Building characteristic dimension/separation	20 / 10	m	fixed linear ellipse inputs
Wind direction	from 270	degree	eastward DMS axis
Design-fire times	1 / 100 / 120	s	terminal output remains on plateau
Time controls (step, maximum, stop)	1 / 1 / 4	s	exact terminal alignment
Dormant ignition	source centre at 5	s	suppresses false first-step stall; occurs after stop
Target CFL / displacement bound	0.10 / 2.0	– / m step^{-1}	all manufactured absolute-speed estimates satisfy the bound
Output cadence	every step	–	preserves the applied heat stimulus
Spread model	WU–E building type 2	–	selects heat-driven urban ROS
Other mechanisms	spotting/crown/stochastic off	–	removes confounders

Figure 1 shows the prescribed fields for the representative equal-energy condition. The bar panel is explicitly a design summary, not simulation output. The predicted output is receiver TOA exactly 1 s and a finite terminal receiver speed recorded when the cell becomes burned on that step. Its magnitude changes continuously with heat and follows the matched head/back/side ellipse-normal mapping. The FTP pair is predicted to differ by a factor of two; the terminal time-stamped heat field is predicted to be zero after reset and is not substituted for the arrival-step stimulus.

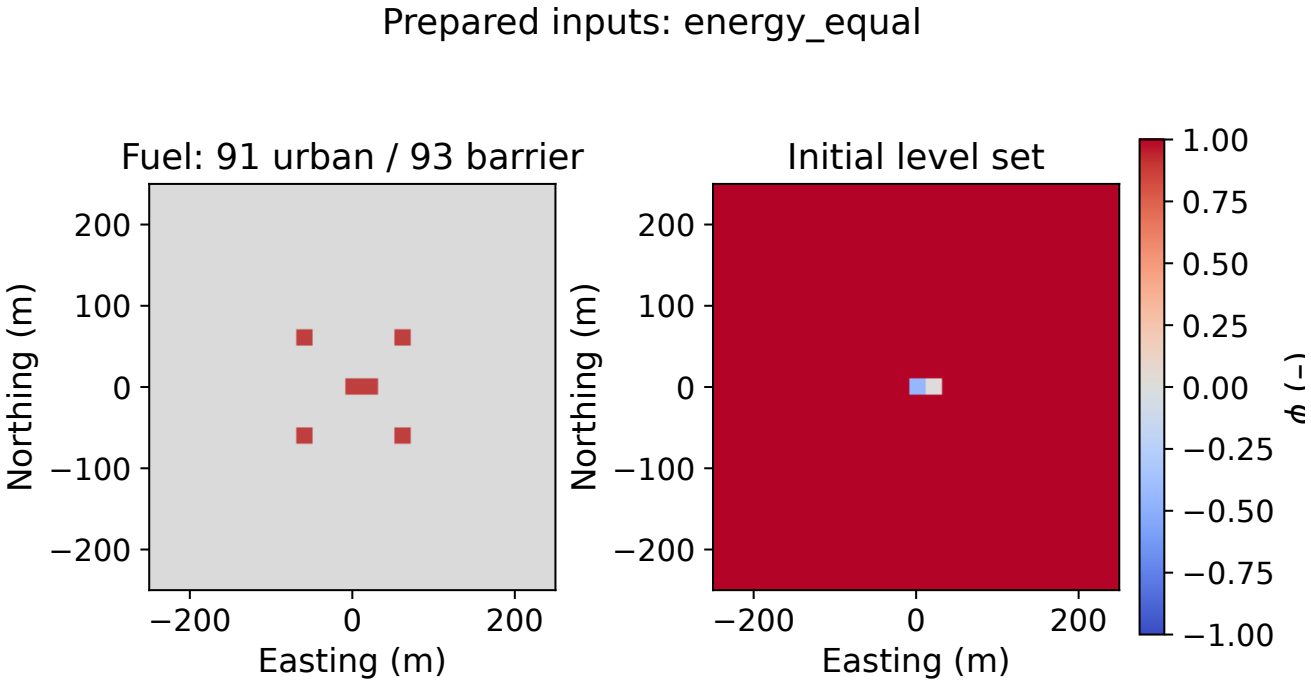


Figure 1: Whole-domain prepared fuel and initial-level-set fields. The separate energy figure shows the designed energy sweep. Coordinates are metres in EPSG:32610; the fields represent the equal-energy condition.

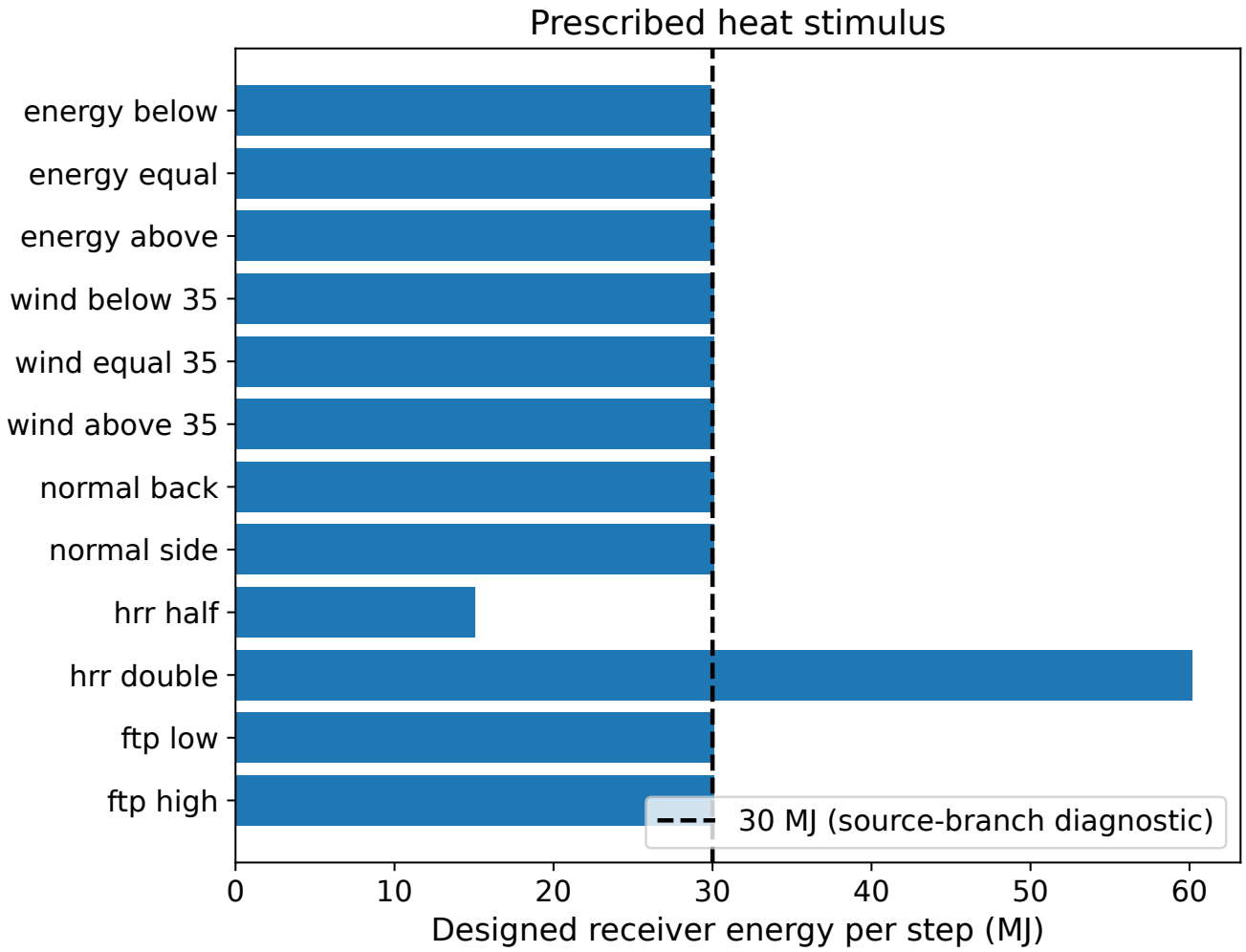


Figure 2: Prescribed receiver energy for all twelve simulation conditions, read from the generated input simulation specification. The dashed line marks the source energy-branch diagnostic, not an acceptance threshold.

6 Actual simulation results

Analysis procedure status is **COMPLETE**; 12 of 12 required simulation conditions have complete black-box evidence. The reported reason is "All required run evidence was complete; every scientific metric passed.". The common model-program SHA-256 identity is `2b58920716af28891d9166c5f1046174214994eb20e820981017e4f2d166b001`; "–" means no completed simulation supplied that evidence. For each completed condition, the analysis retains the actual output times, heat fields at receiver arrival, terminal arrival-time and spread-rate fields, and matching input and reference identities. Previous results are excluded when inputs are regenerated: the scientific state becomes NOT EVALUABLE and the execution state becomes NOT RUN until new evidence is available.

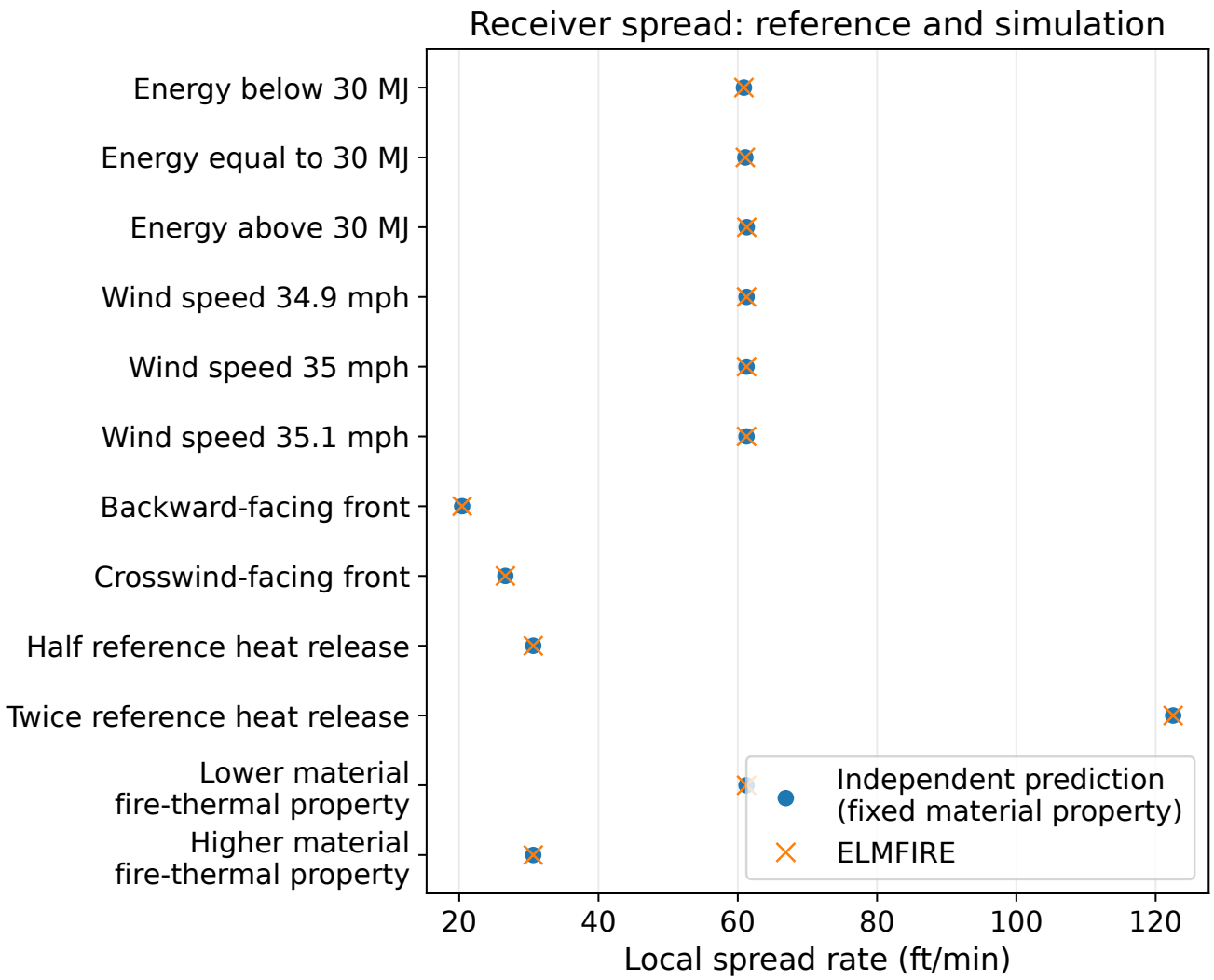


Figure 3: Simulated receiver spread rates and independent predictions for all twelve simulation conditions. The predictions recompute the heat-to-spread relation using the simulated heat stimulus. This comparison isolates the receiver response from differences in the supplied heating; separate figures show the heat stimulus and the spatial result.

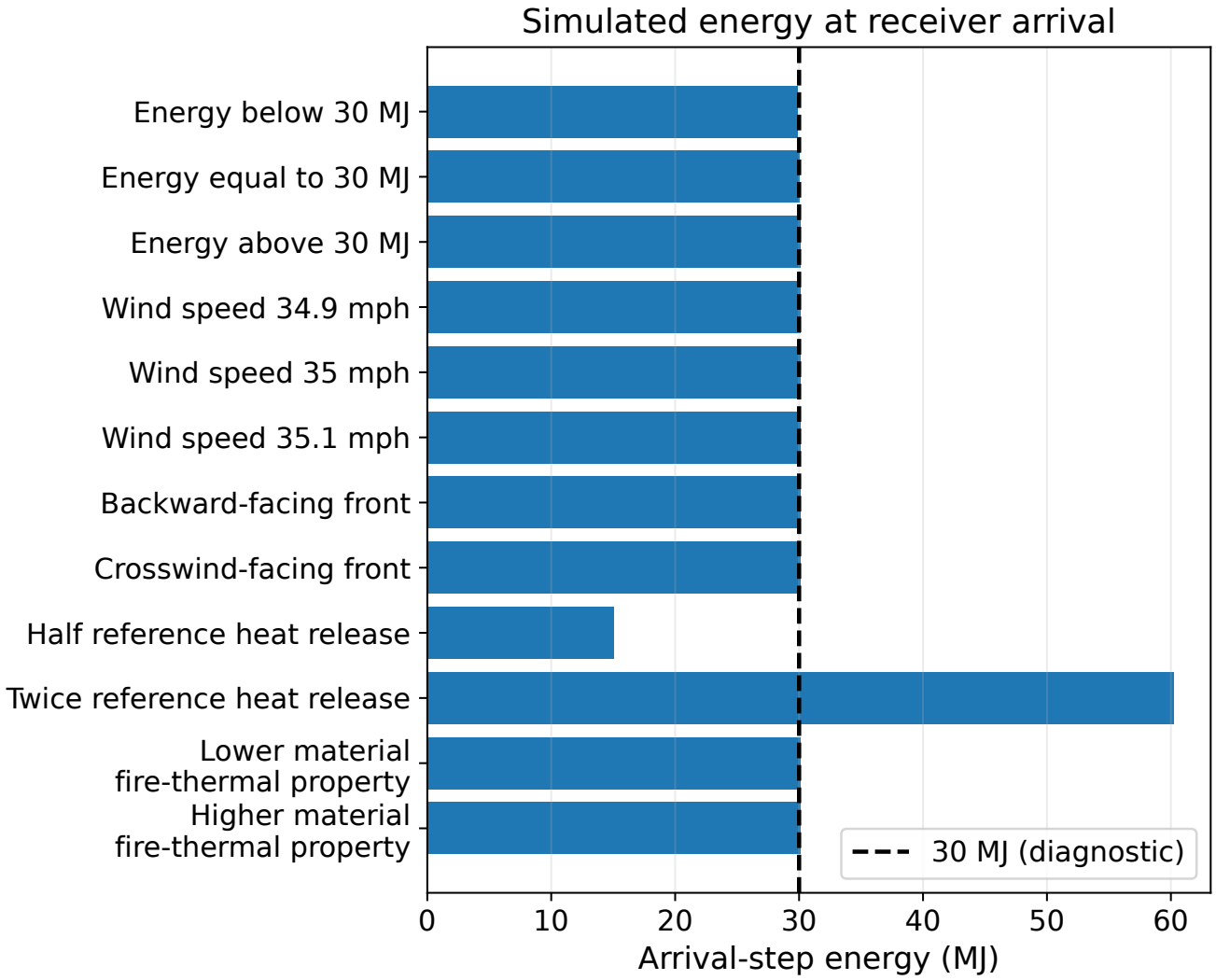


Figure 4: Simulated receiver energy at the selected arrival-step output, calculated from the recorded direct-flame-contact and radiative heat fields. The dashed line identifies the energy-branch transition in the reviewed formulation; it is not an acceptance threshold.

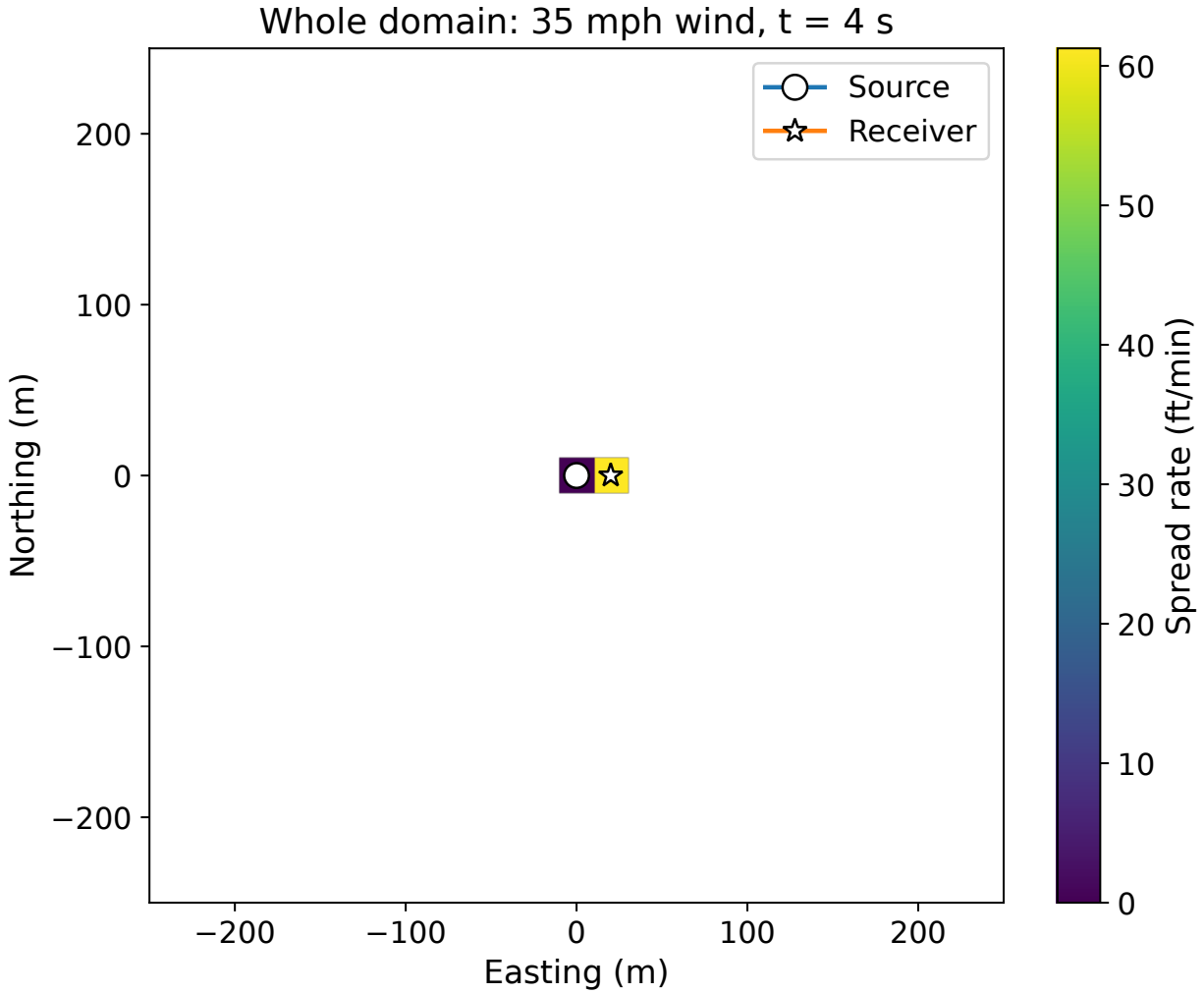


Figure 5: Whole-domain terminal spread rate for 35 mph wind at 4 s, read from the selected ELMFIRE raster. Symbols mark the source and receiver; nodata is blank.

7 Calculated metrics and verification decision

Recorded heat-to-ROS reference calculation relative error — PASS	
Criterion	≤ 0.005
Observed	*Units: fraction.* 0.000207836
Receiver Burned-cell state observability — PASS	
Criterion	TOA = 1 s and terminal VS is finite
Observed	True
Fixed-FTP continuity across 30000 kJ — PASS	
Criterion	normalized speed spread ≤ 0.005
Observed	*Units: fraction.* 4.00432e-08
Wind continuity across 35 mph — PASS	
Criterion	reference calculation error ≤ 0.005
Observed	*Units: fraction.* 5.6595e-08
HRR amplitude response — PASS	
Criterion	double/half heat ratio = 4 +/- 0.04
Observed	*Units: ratio.* 4
Head/back/side response — PASS	
Criterion	matched-wind maximum relative error ≤ 0.005
Observed	*Units: fraction.* 0.000207836
Critical material fire-thermal property inverse response — PASS	
Criterion	speed ratio for 3000/6000 kJ m^-2 = 2 +/- 0.01
Observed	*Units: ratio.* 2

The authoritative case decision sourced from calculated verification and validation metrics is **PASS**. NOT RUN and INCOMPLETE are analysis procedure states; NOT EVALUABLE is their non-passing scientific decision.

8 Technical interpretation

The repaired receiver stencil provides complete, observable arrival and speed evidence. Its positive initial PHI remains positive under the source's input noise, and its side-normal transport no longer competes with an artificial tangential minimum. The observability result therefore separates a usable experiment from agreement with the intended spread law.

The remaining failures identify the heat-to-speed mapping. The source replaces the table FTP with an energy-dependent value and applies a separate wind-speed divisor. The material-pair speeds consequently do not show the required inverse FTP response. The energy and wind comparisons expose the associated branch changes, while the correct HRR amplitude ratio confirms that the prescribed source-amplitude control was realized. See the generated metric rows for the measured errors and decisions; a complete run is not a scientific PASS.

9 Scope and limitations

The case verifies the executable-visible local speed produced by one WU–E urban receiver under a compact deterministic sweep. It does not independently verify the upstream heat-flux model, ignition delay over multiple cells, accumulated exposure across timesteps, building-to-building community spread, or convergence. Internal working variables listed above remain inferred from the reviewed formulation, not observed. All accepted results must share one recorded model-program identity and matching input and reference identities. This consistency does not establish that the program was compiled from the reviewed source revision. The result provides numerical verification, not physical validation.

References

[1] D. M. J. Purnomo et al., "Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model," *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., "Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study," *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.